

KevlarGrid Explicit Solver

Executive Report

Dynamic Mass-Spring Explicit Integration Simulation Summary — Powered by VibeDynaLITE

OUTCOME: PASS (ARRESTED)

1. Impact Simulation & Structural Summary

Fabric Configuration	Striking Projectile
Material preset: Kevlar 29	Initial Mass: 1.7 kg
Young's Modulus: 71.0 GPa	Initial Velocity Vz: 100.0 m/s
Failure Strain Limit: 0.036 m/m	Live Kinetic Energy (KE): 8500.00 Joules
Grid Nodes (Nx × Ny): 100 × 100	Blade Profile Width: 0.1524 m
Grid Spacing dx: 0.0025 m	Edge Thickness: 0.0127 m
Analysis Mode: Mode A (Sizing Multiplier)	
Number of Plies: 20 plies	

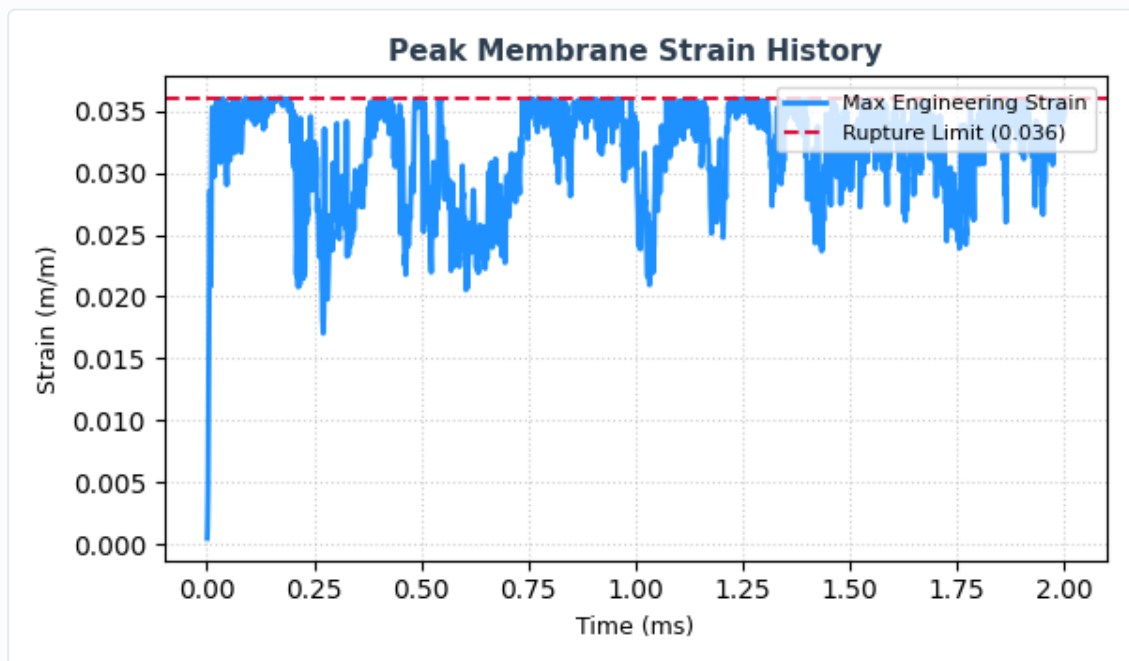
2. Solver Performance Telemetry

Outcome Metric	Value	Standard Limits / Pass Conditions
Projectile Arrest Outcome	PASS (ARRESTED)	MEMBRANE CONTAINMENT SUCCESSFUL
	34731.61 G's	

Outcome Metric	Value	Standard Limits / Pass Conditions
Peak Projectile Deceleration		Dynamic structural loading cap
Yarn Springs Rupture Ratio	7.42%	Fabric structural degradation metric
Residual Velocity	0.00 m/s	Exit velocity post-perforation
Maximum Ply Perforation Index	0 / 20	Total snap-through layer sequence

3. Structural Telemetry Graphs

Peak Fabric Strain Time History



System Energy Balance Conservation

